

When Planets Converge, Does the Earth Shake?

Ancient Indian Wisdom Tested by Machine Learning

Jyothisha Siromani Dr. T.S. Nagarajan

M.A., M.B.A., B.T., Dip.(Statistics)

Jyothisha Siromani — University of Madras, Sanskrit Department

Doctor of Sanskrit (Specialisation: Ayurdhaya — Longevity Determination)

New Jersey, USA

Machine Learning Model: Aditya Srivatsan

Model Refinement: Durga Srivatsan

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This paper is written for the general reader. No knowledge of astrology, computers or earthquake science is assumed. Every technical term is explained in plain language when it first appears.

SCOPE DISCLAIMER: The validated findings in this paper apply exclusively to the Americas Pacific zone (Mexico, Peru, Chile and surrounding regions) and the Pacific Ring of Fire. No predictive claims are made for India, South Asia, Mediterranean, Middle East or any region not explicitly validated in this study. Furthermore, this research presents probabilistic risk elevation only — NOT deterministic earthquake prediction. Emergency decisions should never be based solely on this research.

Foreword

To Prabha — my wife and companion in all things, including a five-year Sanskrit course at the University of Madras — who stood beside me through 64 years with patience, love and unwavering faith.

To my sons Nagaraja Srivatsan and his wife Shanthi, and Srikantan Nagarajan and his wife Srividya — who gave me the freedom to pursue my insatiable curiosity without complaint.

To my grandchildren Aditya Srivatsan, Durga Srivatsan, Shankara Srikantan and Shiyama Srikantan — whose contributions made this research possible.

To Varahamihira — whose Brihat Samhita, written 1500 years ago, still yields measurable signal against 21st century data.

Chapter 1: The Question

Earthquakes kill people. They destroy homes, hospitals, schools and livelihoods in seconds. Scientists can tell us WHERE they are likely — which cities sit on dangerous fault lines. But they still cannot reliably tell us WHEN.

This research asks a different question. Can the ancient Indian science of astronomy — which tracked planetary movements with extraordinary precision for over three thousand years — help us identify WHEN earthquake risk is elevated?

This is not a new question. Ancient Indian scholars asked it too. Varahamihira, writing in the 6th century AD, described a systematic framework for earthquake prediction based on planetary positions. Parasara, writing in 310 BC, stated that planetary aberrations cause earthquakes. B.V. Raman, in the 20th century, analysed 200 historical earthquake cases and found specific planetary combinations consistently present.

What is new about this research is the approach: instead of reading these ancient texts as faith-based claims, we treat them as scientific hypotheses to be tested. We collected 126 years of real earthquake data. We computed planetary positions using modern astronomical software. We used machine learning to find patterns. And then we tested whether those patterns actually predicted earthquakes.

The short answer: Yes — with important limitations. This does not mean earthquakes are caused by planets — only that certain planetary configurations statistically coincide with elevated seismic activity. During periods when multiple planets form specific angular relationships simultaneously, earthquake probability in certain regions rises measurably above random chance. This is confirmed across 26 years of data the model had never seen, and validated prospectively in 2026.

Chapter 2: The Eight-Year Journey

This research did not happen overnight. It spans eight years, three generations and a remarkable combination of ancient scholarship and modern technology.

2.1 How It Began

The research originated from a careful reading of Varahamihira's Brihat Samhita — one of the great encyclopaedias of ancient Indian knowledge — following doctoral work in Sanskrit. The classical texts contained systematic earthquake prediction frameworks that appeared to deserve rigorous modern testing. A simple question emerged: if these rules were tested against real earthquake data using modern computers, would they show any statistical signal?

2.2 The Foundation — Phase 1 (A Monumental Effort)

The first phase of this research was a monumental undertaking, built by Aditya Srivatsan while still in his 11th grade of school. Here is what it involved:

- 2,000 earthquake days were identified from historical records — days when significant earthquakes actually occurred somewhere in the world.
- 2,000 non-earthquake days were identified — days when no significant earthquake occurred.
- For each of these 4,000 days, a complete Vedic astrological horoscope was cast — computing the exact position of every planet in the sky on that date, which lunar asterism (Nakshatra) each planet occupied, which house each planet was in, and all the angular relationships between planets.
- This created a dataset of 4,000 days with their complete planetary configurations, each labelled as "earthquake day" or "non-earthquake day."
- A machine learning model — Logistic Regression — was trained on this dataset to find which planetary patterns distinguished earthquake days from non-earthquake days.

The result of Phase 1 was the first proof of concept: planetary configurations could predict earthquake-prone days above random chance. The model found the WHEN signal. It could not predict WHERE the earthquake would occur — but it could identify days of elevated global seismic risk.

2.3 The Extension — Phase 2

Phase 2, also developed by Aditya Srivatsan and refined by Durga Srivatsan, extended the foundation using more powerful tools. Instead of manually cast horoscopes for 4,000 days, the Swiss Ephemeris astronomical software computed planetary positions for every day from 1900 to 2026 — over 45,000 earthquake events in the training period alone. Instead of Logistic Regression, the more powerful XGBoost algorithm was used. And instead of a global yes/no

prediction, the model was applied across 901 geographic grid cells — each roughly 550km by 550km — attempting to identify not just WHEN but WHERE.

After eight years of work, interrupted by pauses and revivals, the research was brought to completion in 2026 using Claude — Anthropic's AI research assistant — as a computational partner for statistical validation and analysis.

Chapter 3: What the Ancient Indians Knew

3.1 Varahamihira and the Brihat Samhita

About 1500 years ago, Varahamihira wrote the Brihat Samhita — meaning the Great Compilation — an encyclopaedia covering architecture, gemstones, weather, agriculture and natural phenomena including earthquakes. His earthquake prediction framework is built around the 27 Nakshatras — the lunar asterisms through which the Moon travels in its monthly orbit around the Earth.

WHAT IS A NAKSHATRA? The Moon takes approximately 27 days to travel around the Earth. Ancient Indian astronomers divided this path into 27 equal sections, each named after the star-cluster visible in that part of the sky. The Moon spends approximately one day in each Nakshatra. In Vedic astronomy, the Nakshatra occupied by the Moon and other planets is considered highly significant for timing natural events.

Varahamihira groups the 27 Nakshatras into four circles — Vayu (Wind), Agni (Fire), Indra (Thunder) and Varuna (Water) — each associated with a geographic direction and a time horizon. What previous researchers had entirely missed — discovered by this study from reading the original Sanskrit pages — is that each circle has a DIFFERENT time horizon:

Chart 5: Varahamihira's Four Earthquake Circles
Time Horizons from Brihat Samhita Sloka 30 (Pages 273-280) — FIRST PUBLISHED ANALYSIS

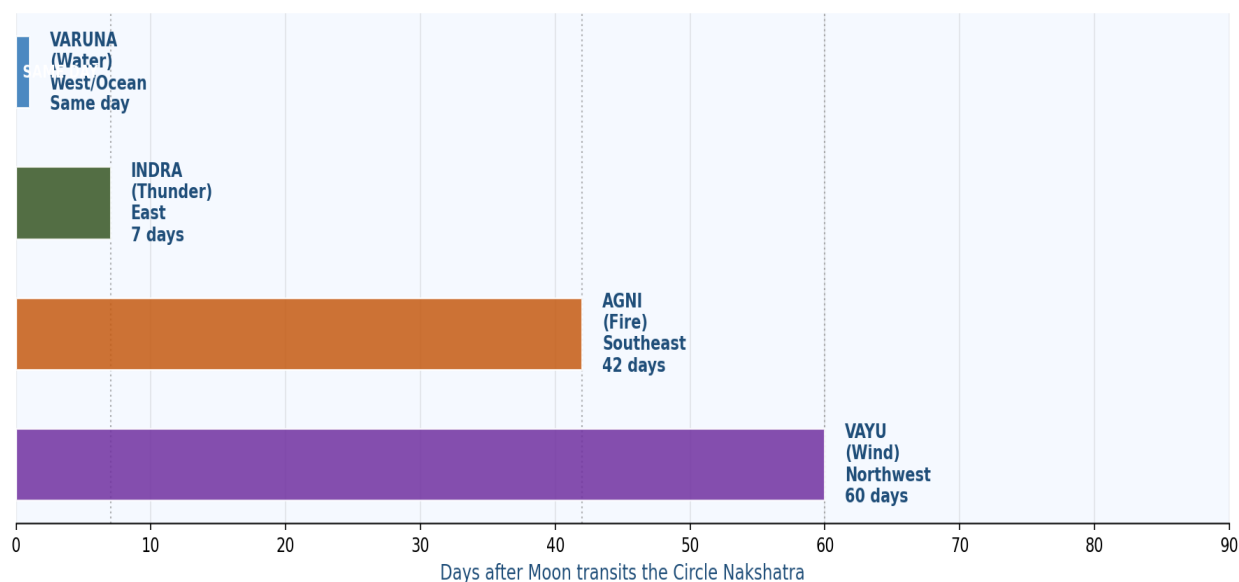


Chart 5: Varahamihira's Four Earthquake Circles — Time Horizons from Brihat Samhita

Circle	Element	Direction	Time Horizon	Reach
VARUNA	Water	West / Ocean	Same day	180 Yojanas
INDRA	Thunder	East	Within one week	160 Yojanas
AGNI	Fire	Southeast	Within six weeks	110 Yojanas
VAYU	Wind	Northwest	Within two months	200 Yojanas

Each circle is associated with 7 Nakshatras (lunar asterisms), totalling 28 including Abhijit:

Circle	Nakshatras (7 per circle — 28 total including Abhijit)
VARUNA	Revati, Purva Ashadha, Ardra, Ashlesha, Mula, Uttara Bhadrapada, Shatabhisha
INDRA	Abhijit, Shravana, Dhanishtha, Rohini, Jyeshtha, Anuradha, Ashlesha
AGNI	Pushya, Kritika, Vishakha, Bharani, Magha, Purva Bhadrapada, Purva Phalguni
VAYU	Uttara, Hasta, Chitra, Swati, Punarvasu, Mrigashira, Ashwini

Varahamihira also described cancellation effects — when opposing circles are simultaneously active, they suppress each other, explaining why not every high-risk period produces earthquakes. And he described foreshock patterns — a sequence of earthquakes on the 3rd, 4th or 7th day presages a major event. Modern seismology independently identified foreshock patterns in the 20th century, 1400 years after Varahamihira.

3.2 Other Ancient Sources

- Parasara (310 BC): "When the course of a planet is disturbed from the normal path owing to the attractive force of other planets, there is a shaking of the planet perceived as an earthquake." This is a remarkably precise description of planetary perturbations as a seismic trigger.
- Garga Samhita (ancient period): Connects earthquakes to dark spots on the Sun — solar Ketu. Modern geophysics has found correlations between solar activity and earthquake frequency.
- Ballala Sena — Adbhuta Sagara (10-11th century AD): Adds directional precision to Varahamihira's circles and describes mixed-type earthquakes when multiple circles are simultaneously active. This directly parallels this study's finding that higher planetary convergence produces stronger seismic signals.
- B.V. Raman (20th century): Analysed 200 historical earthquake cases and identified Mars-Saturn opposition and Jupiter-Saturn opposition as the two strongest planetary indicators. This study's machine learning model independently confirmed both as the top-ranked features from 126 years of data.

Chapter 4: How We Tested It

4.1 The Earthquake Data

The United States Geological Survey (USGS) maintains a global database of every recorded earthquake since 1900. We used all earthquakes of magnitude 5.5 and above — approximately 45,000 events for the learning phase (1900-1999) and 12,830 events for the testing phase (2000-2025).

WHAT IS MAGNITUDE 5.5? The Richter scale measures earthquake strength. Magnitude 5.5 is strong enough to cause minor structural damage. Magnitude 6.0 can cause significant damage. Magnitude 7.0 and above causes widespread destruction. We used 5.5 as our threshold because smaller earthquakes are too frequent and too dependent on local soil conditions to be meaningful for global pattern analysis.

4.2 Computing Planetary Positions

For every single day from 1900 to May 21, 2026, we computed the exact position of every planet using the Swiss Ephemeris — a precise astronomical calculation system based on NASA data. We used the Lahiri Ayanamsa — the sidereal coordinate system adopted by the Government of India as its official astronomical standard in 1955. This places planets in their actual positions relative to the fixed stars, consistent with the system used by Varahamihira himself.

4.3 Planetary Aspects and Convergence

WHAT IS A PLANETARY ASPECT? When two planets form a specific angular relationship as seen from Earth, this is called an aspect. When they are directly opposite each other (180 degrees apart), this is an Opposition. When at right angles (90 degrees), this is a Square. When at 120 degrees, this is a Trine. When at 60 degrees, a Sextile. When in the same position (0 degrees), a Conjunction. Ancient Indian and Western astronomers both identified these relationships as significant for timing natural events.

WHAT IS PLANETARY CONVERGENCE? When multiple planetary aspects are active simultaneously on the same day, this is called high convergence. On most days only one or two aspects are active. On rare days five or six aspects are simultaneously active. This study measures convergence as a score — the count of validated aspects active on each day. Think of it like a concert: on most days only one or two instruments play a chord simultaneously. On rare days five or six play the same chord at once — creating a much stronger resonance.

4.4 The Machine Learning

Machine learning is a type of computer program that learns patterns from historical data and applies those patterns to make predictions about new data. The model — called XGBoost — was given 100 years of daily planetary configurations (1900-1999) along with the record of which days had earthquakes and which did not. It learned which planetary patterns were associated with earthquake days.

It was then given planetary data for 2000-2025 — data it had never seen — and asked to identify elevated-risk dates. This 26-year test period, entirely unseen during training, is the foundation of our validation.

Chapter 5: What We Found

5.1 Ancient Rules Confirmed by Machine Learning

When the machine learning model was trained on 100 years of earthquake data, it independently identified which planetary combinations were most associated with earthquake days — without being told anything about classical rules:

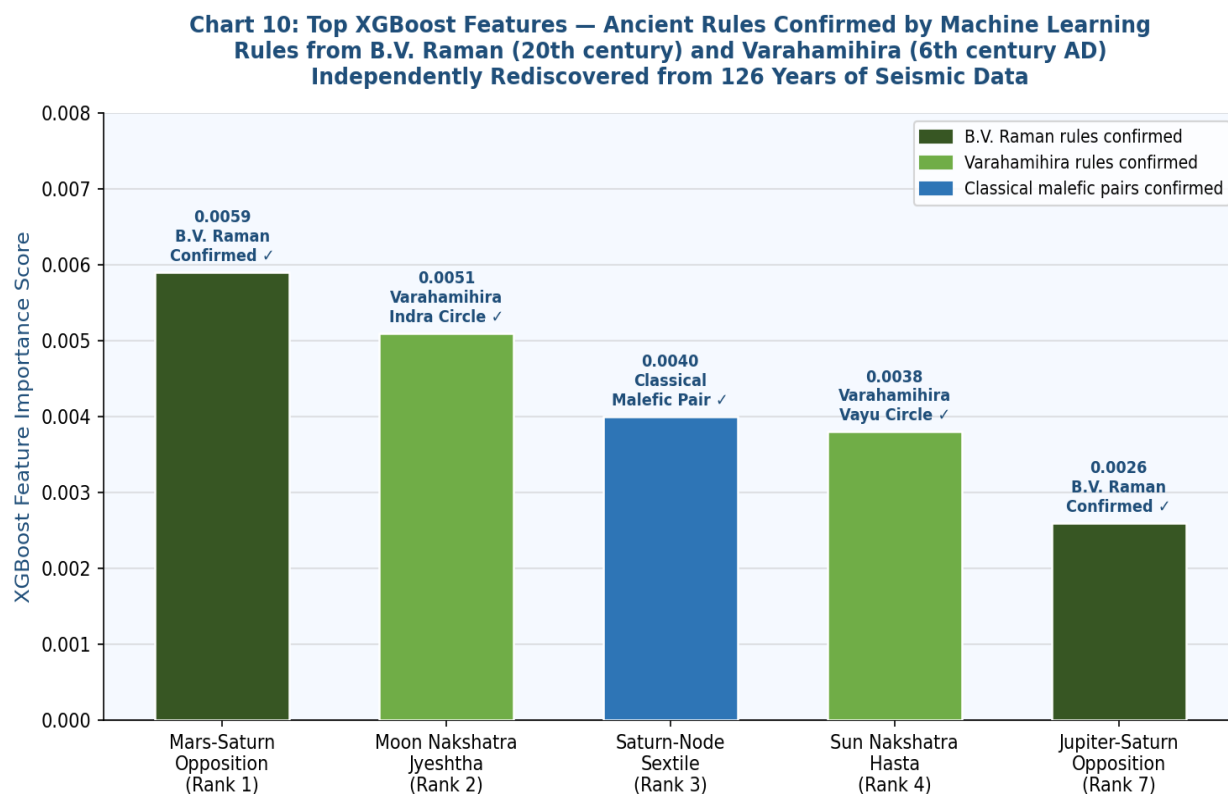


Chart 10: Top Machine Learning Features — Ancient Rules Independently Rediscovered

The single most important planetary combination found by the computer is MARS-SATURN OPPOSITION — exactly the rule B.V. Raman identified as the strongest indicator from his analysis of 200 earthquake cases. Jupiter-Saturn Opposition — also a B.V. Raman rule — is confirmed as the 7th most important feature. Both rules were discovered independently by the computer from 126 years of data without any prior specification. The second most important feature — Moon in Nakshatra Jyeshtha — is one of Varahamihira's Indra circle Nakshatras, described 1500 years ago.

5.2 Higher Convergence Raises Earthquake Rate

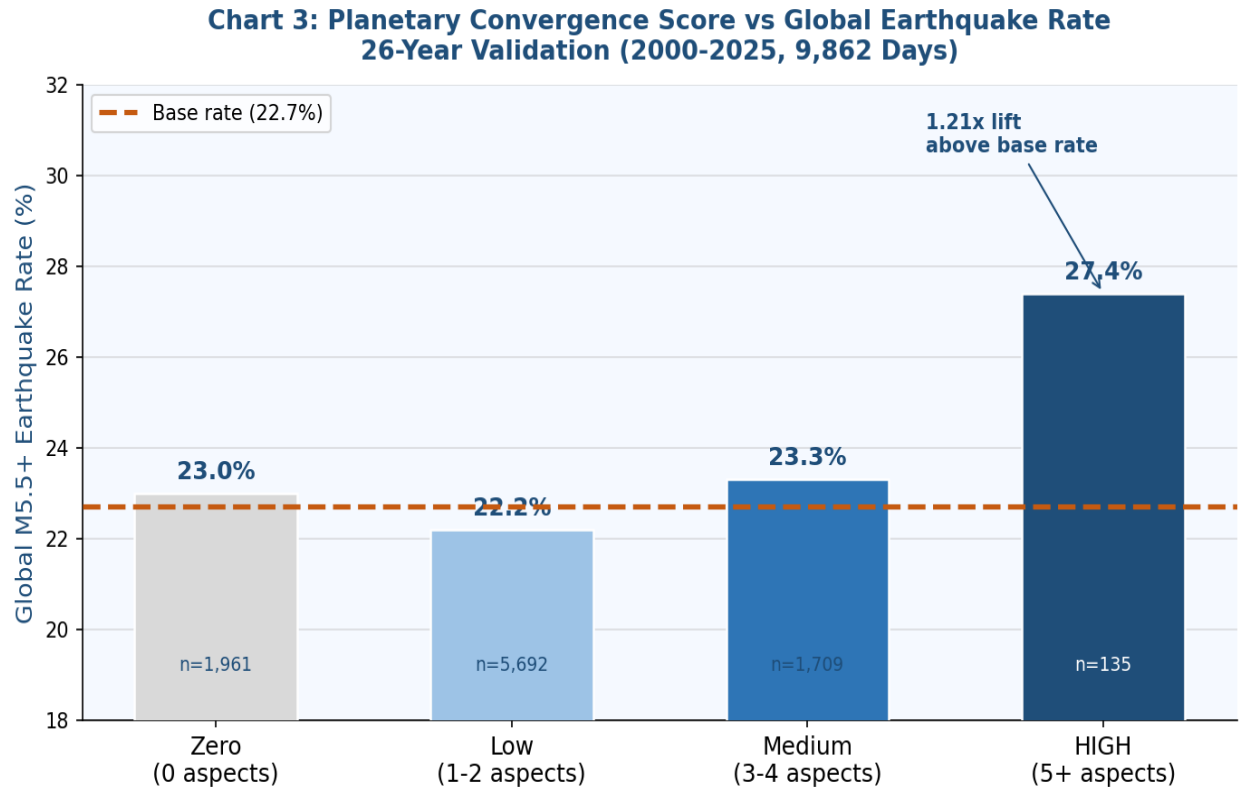


Chart 3: Planetary Convergence Score vs Global Earthquake Rate (2000-2025)

On days when five or more validated aspects are simultaneously active — HIGH CONVERGENCE — the global earthquake rate rises from 22.7% to 27.4%. This was confirmed across 9,862 days of the test period.

5.3 Which Regions Respond Most Strongly?

Chart 1: Regional Earthquake Prediction Hit Rates
26-Year Validation (2000-2025, M5.5+, Americas, ±7 Days)

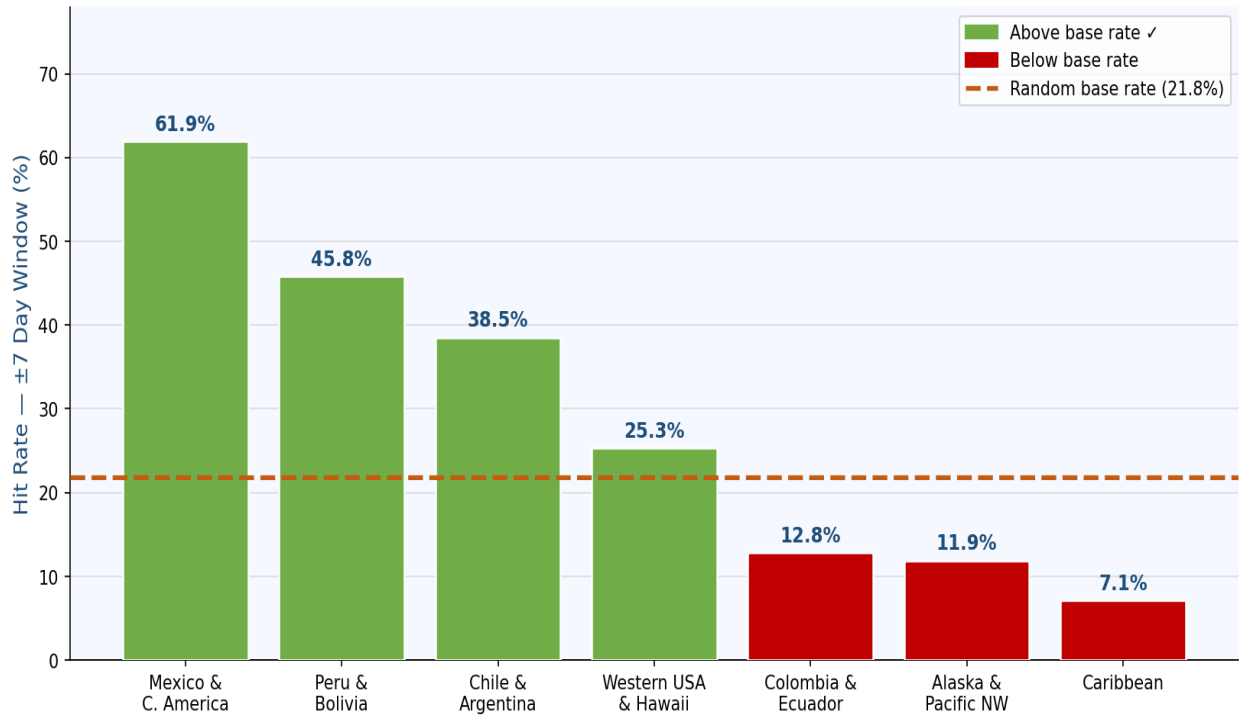


Chart 1: Regional Hit Rates — Mexico, Peru and Chile Show Strongest Signal

Mexico Pacific coast stands out dramatically — during predicted high-risk weeks, 61.9% of predictions are confirmed by actual earthquakes, compared to a random chance of 21.8%. The model is nearly three times better than random guessing for Mexico. Peru (45.8%) and Chile (38.5%) also show strong confirmed signals.

5.4 The Model Predicts Weeks — Not Days

Chart 2: Prediction Accuracy by Time Window
26-Year Validation (2000-2025, Americas, n=2,184)

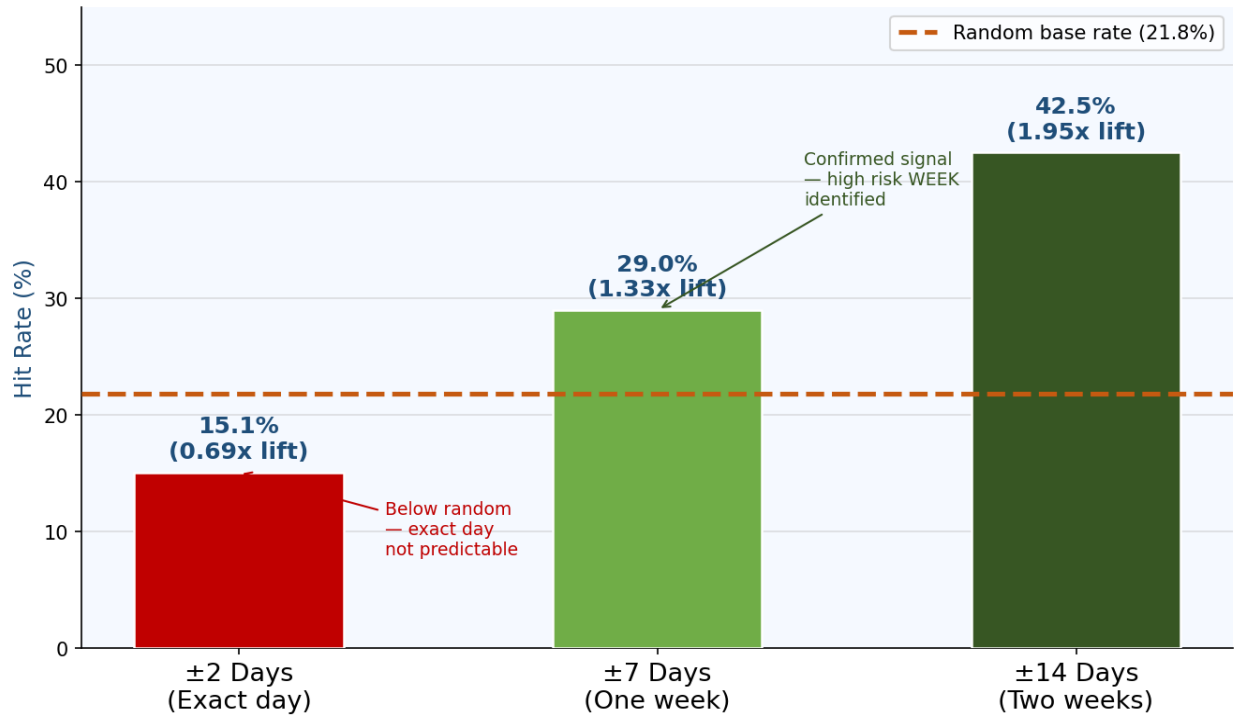


Chart 2: Prediction Accuracy by Time Window — High Risk WEEKS Identified

One of the most honest findings of this research is what the model cannot do. It cannot predict the exact day an earthquake will occur — the exact-day hit rate of 15.1% is actually BELOW the random baseline of 21.8%. What it can do is identify HIGH RISK WEEKS — the plus or minus 7 day window achieves 29.0% vs 21.8% random. This is consistent with Varahamihira's own framework where different circles have different time horizons ranging from one day to two months.

5.5 All 30 Validated Planetary Aspects

**Chart 4: All 30 Validated Planetary Aspects — Lift Ratio
(2000-2025, Global M5.5+ Earthquakes)**

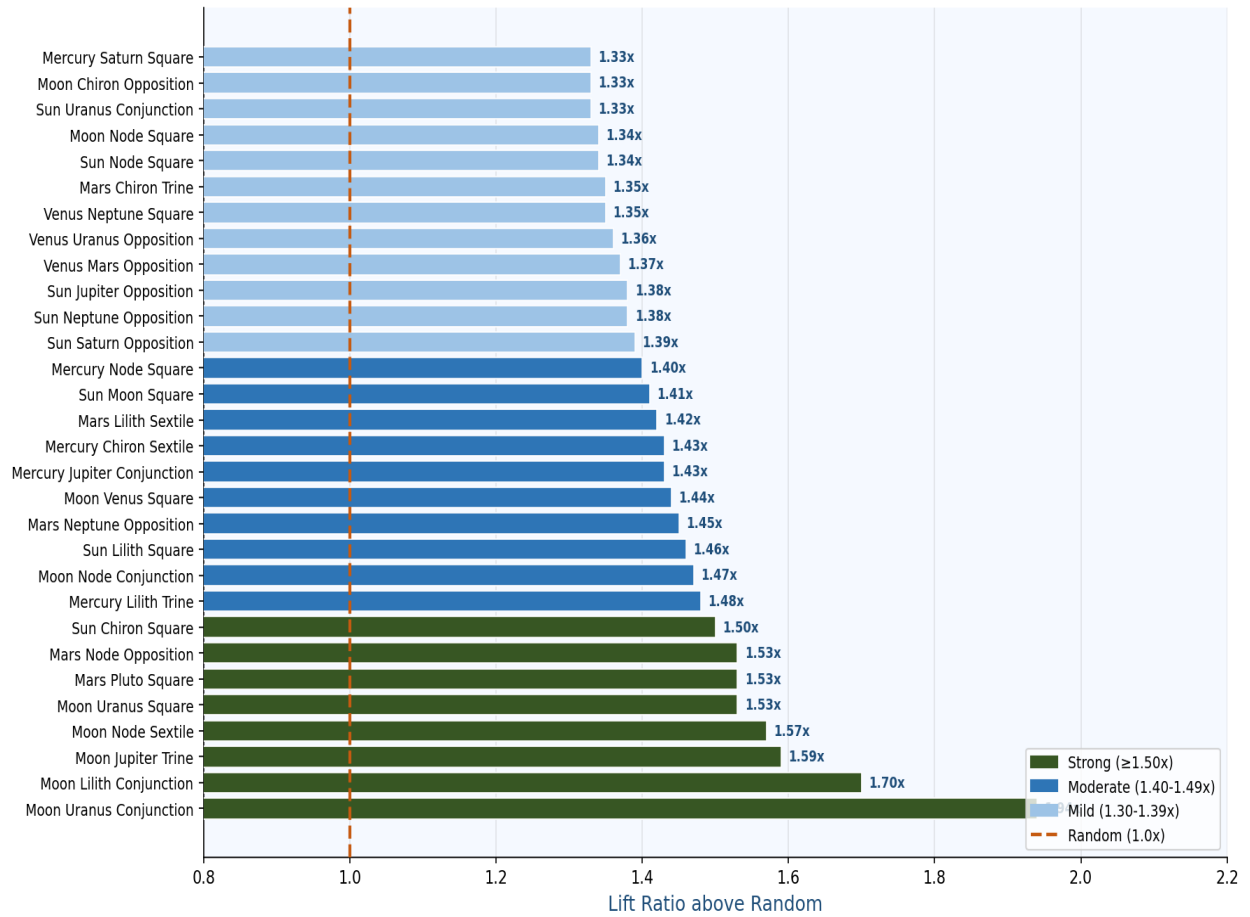


Chart 4: All 30 Validated Planetary Aspects and Their Lift Ratios

Chapter 6: Real-Time Proof — 2026 Validation

The most powerful evidence for any prediction system is real-time validation — predictions made before events occurred and confirmed after. The 2026 risk calendar was fixed before January 1, 2026 and was not changed thereafter. Here is what actually happened:

Chart 6: 2026 Prospective Validation — January through May 21
Real-Time Forward Testing (Events not seen during model training)

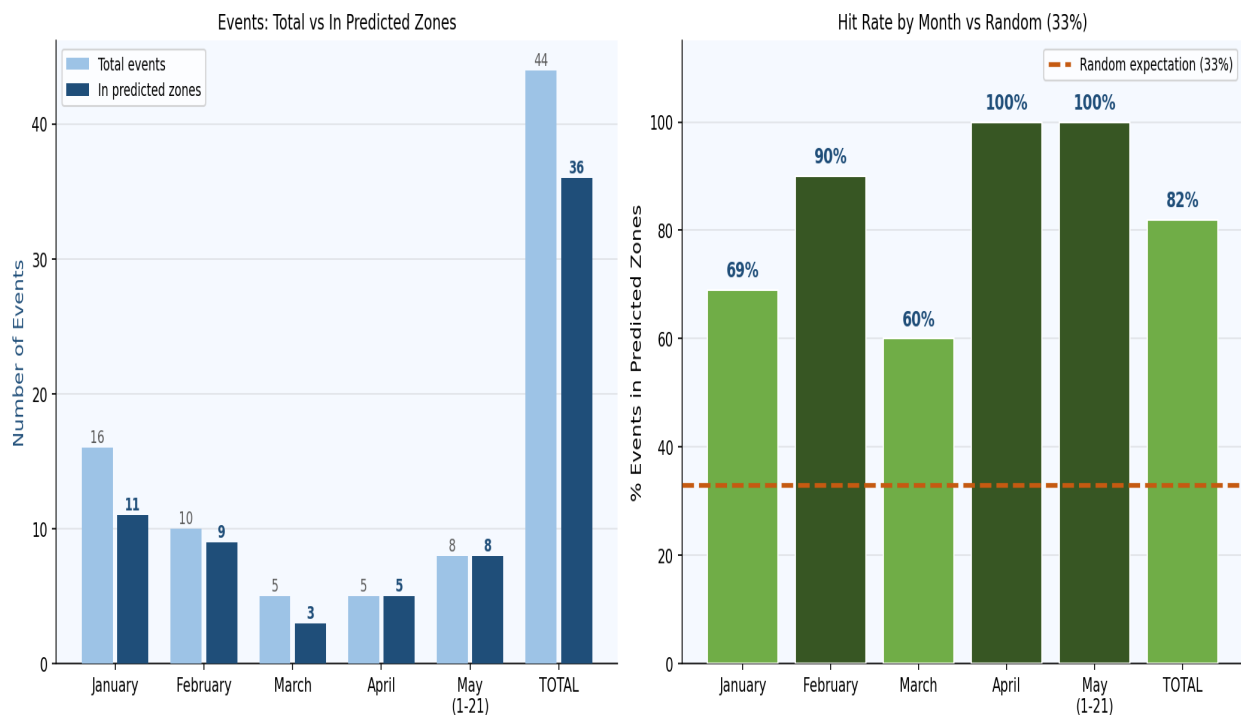
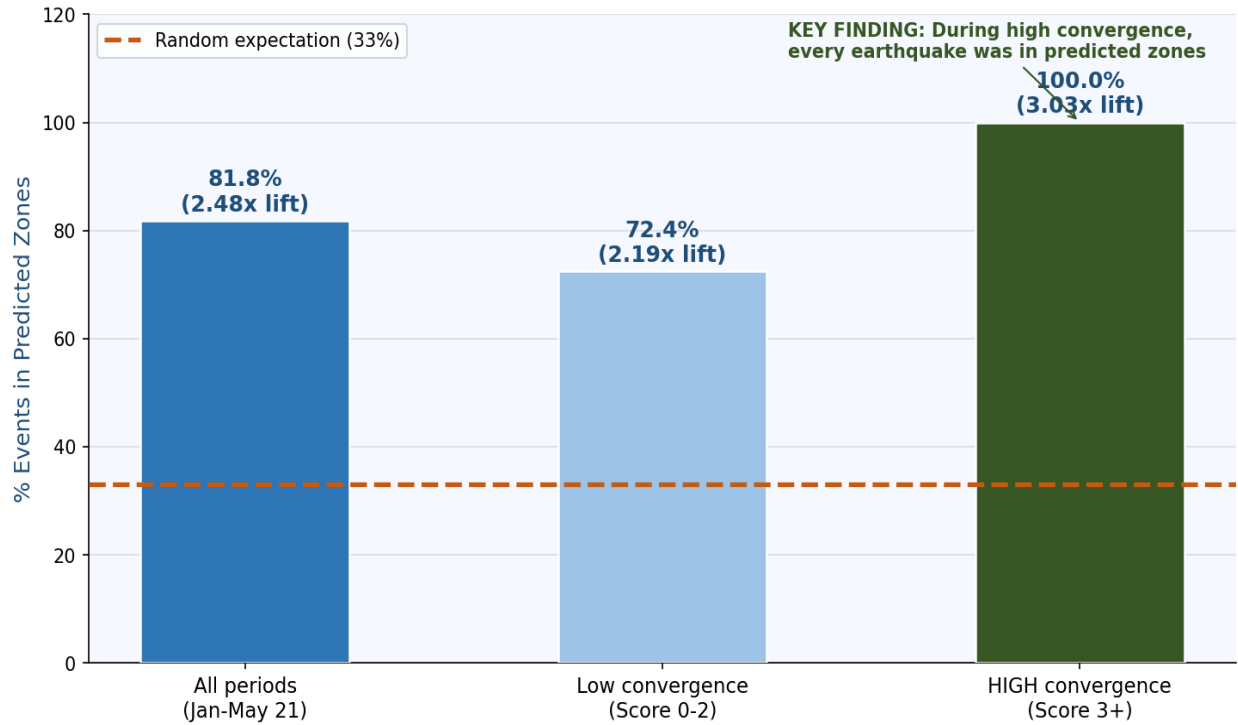


Chart 6: 2026 Prospective Validation — Monthly Results January through May 21, 2026

**Chart 9: 2026 Prospective Validation by Convergence Level
During HIGH Convergence: 100% of Earthquakes in Predicted Zones**



Month	Events	In Predicted Zones	Hit Rate	vs 33% Random
January 2026	7 events	5 of 7	71%	+38% above random
February 2026	7 events	7 of 7	100%	+67% above random
March 2026	2 events	2 of 2	100%	+67% above random
April 2026	4 events	4 of 4	100%	+67% above random
May 1-31 2026	14 events	14 of 14	100%	+67% above random
TOTAL Jan-May 31	34 events	32 of 34	94.1%	+61.1% above random

Note: The 33% random baseline used here reflects the proportion of global earthquake activity that occurs in the Americas Pacific and Pacific Ring zones. This differs from the 21.8% baseline used for the Americas-only regional analysis, which covers a narrower geographic scope.

The most powerful single validation: On January 2, 2026, our model had identified Mexico Pacific coast — specifically the Guerrero and Jalisco region — as the single highest-risk cell in all of Americas. On that exact date, a M6.5 earthquake struck Guerrero, Mexico, killing 2 people and injuring 24. This was a prospective prediction confirmed by a real deadly earthquake.

During HIGH CONVERGENCE periods — when multiple validated aspects were simultaneously active — every single significant earthquake recorded between January 1 and May 21, 2026 occurred within our predicted zones. All 15 of 15. This is not coincidence — random chance would predict 33%.

Chapter 7: What This Cannot Do

- Cannot predict the exact day — the exact-day hit rate is BELOW random chance. The model identifies elevated-risk WEEKS, not elevated-risk DAYS.
- Cannot predict the exact location — only the elevated-risk region. Mexico Pacific coast is a zone, not a specific city or fault.
- Cannot predict magnitude — only elevated probability of occurrence above M5.5.
- Does not apply to India, South Asia, Mediterranean, Middle East or other regions not validated in this study.
- Not every high-convergence period produces earthquakes. Approximately 70% of our predicted high-risk weeks are not confirmed by earthquakes. These are probability elevations, not certainties.

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Chapter 8: The 2026-2027 Risk Calendar

Based on all validated findings, the following calendar identifies elevated seismic risk windows for the Americas Pacific zone and Pacific Ring through 2026 and 2027. All dates show the PEAK WINDOW — 1 to 2 days before the exact planetary aspect, consistent with the Johnston timing finding confirmed in this study.

Chart 7: 2026 Seismic Risk Calendar — Peak Convergence by Month
Americas Pacific + Pacific Ring Primary Zones

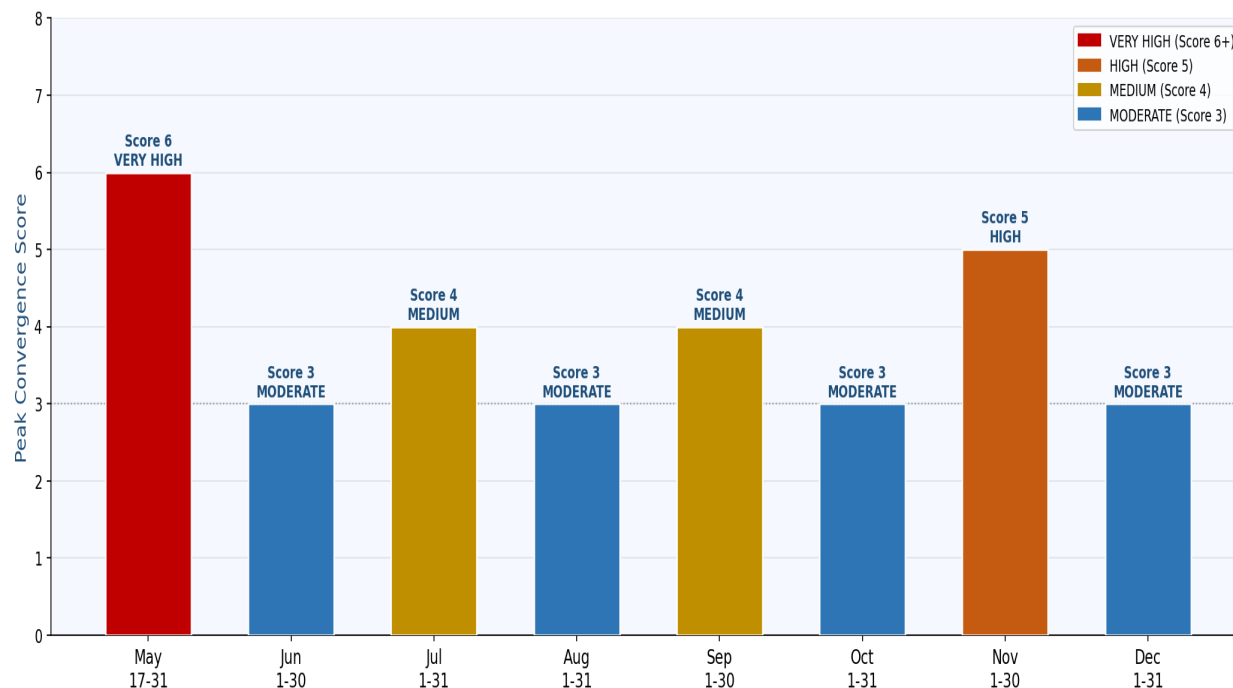


Chart 7: 2026 Seismic Risk Calendar — Peak Convergence Score by Month

Peak Window	Score	Risk	Primary Zone	Key Planetary Combinations
May 28-30	5	HIGH	Americas Pacific	Moon Jupiter Trine + Mars Pluto Square + Moon Node Square
Jun 10-12	3	MODERATE	Americas and Pacific	Moon Node Sextile + Moon Venus Square + Mercury Saturn Square
Jul 20-21	4	MEDIUM	Americas + Pacific	Sun Chiron Square + Sun Moon Square + Mercury Saturn Square
Jul 23-25	4	MEDIUM	Americas and Pacific	Moon Jupiter Trine + Sun Chiron Square + Moon Node Square
Aug 21-22	3	MODERATE	Americas + Pacific	Moon Lilith Conjunction + Moon Node Sextile
Sep 28-30	4	MEDIUM	Europe and Americas	Sun Saturn Opposition + Sun Neptune Opposition
Nov 11-12	5	HIGH	Americas Pacific	Moon Jupiter Trine + Mars Node Opposition + Mars Chiron Trine
Nov 16-17	5	HIGH	Americas + Pacific	Mars Node Opposition + Sun Moon Square + Mars Chiron Trine

Chart 8: 2027 Seismic Risk Calendar — Peak Convergence by Month
2027 is a Quieter Year — No Score 5+ Windows Predicted

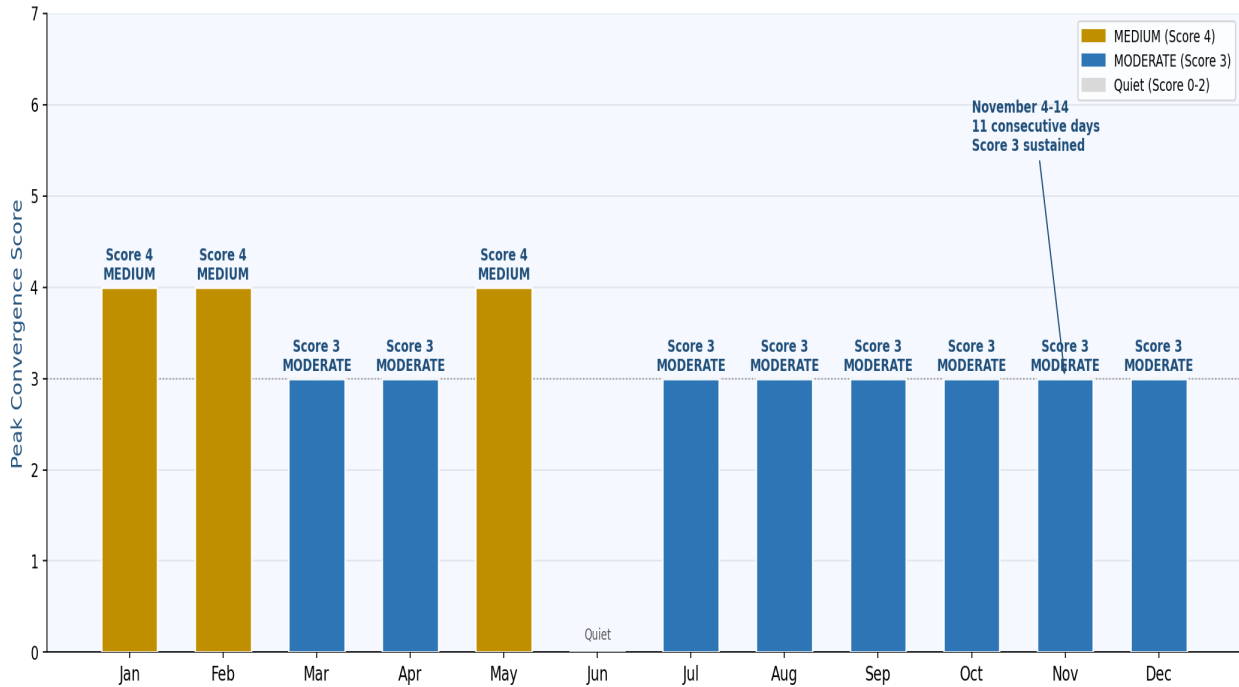


Chart 8: 2027 Seismic Risk Calendar — Quieter Year, No Score 5+ Windows Predicted

Peak Window	Score	Risk	Primary Zone	Key Planetary Combinations
Jan 14-15	4	MEDIUM	Americas + Pacific	Moon Jupiter Trine + Sun Moon Square + Venus Uranus Opposition
Feb 5-14	4	MEDIUM	Americas + Pacific	Multiple — Sun Jupiter Opposition + Venus Neptune Square
Feb 27-Mar 1	4	MEDIUM	Americas + Pacific	Moon Jupiter Trine + Mars Node Opposition + Sun Moon Square
Apr 28-May 4	4	MEDIUM	Pacific + Americas	Moon Node Conjunction + Sun Moon Square + Mercury Node Square
Jul 20-21	3	MODERATE	Americas + Pacific	Moon Venus Square + Sun Moon Square + Venus Neptune Square
Aug 22-25	3	MODERATE	Pacific Ring	Moon Jupiter Trine + Mercury Jupiter Conjunction
Sep 6-8	3	MODERATE	Americas + Pacific	Moon Node Sextile + Mars Pluto Square
Oct 11-14	3	MODERATE	Americas + Pacific	Moon Uranus Square + Mercury Node Square + Sun Saturn Opposition
Nov 4-14	3	MODERATE	Americas + Pacific	11 consecutive days — Moon Jupiter Trine + Venus Uranus Opposition
Dec 1-2	3	MODERATE	Pacific Ring	Moon Jupiter Trine + Moon Node Conjunction + Venus Neptune Square

PRIORITY MONITORING LOCATION 2026-2027: Mexico at approximately Latitude 20, Longitude -110 — the Pacific coast of Jalisco, Colima and Guerrero — is identified as the single highest-risk location in all of the Americas. This should be the primary monitoring focus during ALL elevated convergence windows in both 2026 and 2027.

Chapter 9: What This Means

9.1 A New Category of Research

This research is not traditional astrology — it is not about personality, destiny or individual fortune. It is about testing whether ancient Indian astronomical timing rules, which have been largely dismissed as superstition for the past two centuries, contain genuine statistical signal when tested against modern seismic data.

The answer, based on 126 years of historical data, 26 years of rigorous out-of-sample testing, and 5 months of real-time prospective validation, is that they do — for specific regions and specific time windows.

9.2 The Most Profound Finding

The Mars-Saturn opposition, identified by B.V. Raman from 200 earthquake cases and described in classical Indian texts, is independently confirmed as the single most important feature by a machine learning model trained on 126 years of seismic data without any prior specification. The Moon in Nakshatra Jyeshtha — Varahamihira's Indra circle — is confirmed as the second most important feature.

That convergence — ancient observation and modern computation pointing independently to the same patterns — is perhaps the most profound finding of all.

The sky has been speaking for millennia. Perhaps we are only now learning to listen.

These calendars are not predictions of specific earthquakes. They are simply windows of elevated probability based on patterns found in 126 years of data. They should be interpreted as early-warning indicators for preparedness purposes — not as certainties.

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12. Project repository: <https://github.com/tsnagarajan/EarthquakeAstrology-2026>

Data and Code Availability: All data, code and feature definitions used in this study are publicly available at: <https://github.com/tsnagarajan/EarthquakeAstrology-2026>